

Unit 10

GROUP 2

27 items · 70 marks

Syllabus sections in this unit

10.1 Similarities and trends in the properties of the Group 2 metals, magnesium to barium, and their compounds

Unit 10: Group 2**1. M/J 26/P24/Q3(c)**

(c) (i) Describe what is observed when white solid strontium nitrate decomposes on heating.

[1]

(ii) State the trend in solubility down the Group 2 hydroxides.

[1]

2. MAR 26/P22/Q1(b)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(b) When strontium reacts with water, an alkaline solution is produced.

(i) Write an equation for this reaction. Include state symbols.

[2]

(ii) Samples of the alkaline solution produced in 1(b)(i) are tested with dilute HCl(aq) and separately with dilute $\text{H}_2\text{SO}_4\text{(aq)}$.

State what you would observe in each test.

observation with dilute HCl(aq)

observation with dilute $\text{H}_2\text{SO}_4\text{(aq)}$

[2]

3. MAR 26/P22/Q1(c)

Context

1 The Group 2 elements Mg to Ba show trends in their physical properties down the group.

(c) The nitrates and carbonates of Group 2 elements decompose on heating.

(i) State the trend in the thermal stability of the Group 2 carbonates down the group.

[1]

(ii) Write an equation for the thermal decomposition of $\text{Ca(NO}_3)_2$.

[1]

4. O/N 25/P21/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

5. O/N 25/P22/Q2(f)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(f) Aqueous MgCl_2 reacts with aqueous Na_2CO_3 to form a white precipitate. Upon heating, the white precipitate undergoes thermal decomposition.

(i) Construct an equation to show the reaction of aqueous MgCl_2 with aqueous Na_2CO_3 . Use state symbols in your equation.

[2]

(ii) Identify the products of the thermal decomposition of the white precipitate.

[1]

(iii) State the trend in thermal stability of the Group 2 carbonates down the group.

[1]

6. O/N 25/P23/Q2(d)

Context

2 The Period 3 elements show trends in physical and chemical properties across the period.

(d) Group 2 nitrates decompose on heating to form oxides.

(i) State the trend in thermal stability of the Group 2 nitrates down the group.

[1]

(ii) Identify the other products of the thermal decomposition of Group 2 nitrates.

[1]

7. MAR 25/P22/Q2(d)

(d) (i) State what is observed when dilute hydrochloric acid is added to separate samples of barium oxide and barium carbonate.

barium oxide

barium carbonate

[2]

(ii) Write an equation for the reaction of strontium, Sr, with an excess of cold water.

[1]

(iii) State the variation in solubility of the Group 2 sulfates down the group.

[1]

8. O/N 24/P21/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute $\text{HCl}(\text{aq})$.

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

9. O/N 24/P22/Q3(a)(iv)-(v)

Context

3 Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

Context: part (a) (introduction)

(a) Nitrogen is found in inorganic compounds such as nitrogen oxides (NO_x), nitrates and nitric acid.

(iv) Nitric acid reacts with basic oxides to form nitrates.

Write an equation for the reaction of nitric acid with calcium oxide.

[1]

(v) Describe what is seen when solid calcium nitrate is heated strongly.

[1]

10. O/N 24/P23/Q3(b)(i)-(ii)

Context

3 The Group 14 elements show a change from non-metallic to metallic character down the group.

Context: part (b) (introduction)

(b) Carbon is found in inorganic compounds such as carbonates.

(i) Write an equation for the reaction of magnesium carbonate with dilute $\text{HCl}(\text{aq})$.

[1]

(ii) Describe the thermal stability of the carbonates down Group 2.

[1]

11. M/J 24/P22/Q2(c)

Context

2 Separate samples of Na_2CO_3 and NaHCO_3 react with $\text{HCl}(\text{aq})$ to produce the same products, as shown in Table 2.1.

Table 2.1

reaction	equation	$\Delta H/\text{kJ mol}^{-1}$
1	$\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	ΔH_1
2	$\text{NaHCO}_3 + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$	$\Delta H_2 = +27.2$

(c) **Z** is a salt that contains a Period 4 element from Group 2. When **Z** is heated brown gas forms.

Identify the formula of **Z** and use it to write an equation for the reaction.

[2]

12. M/J 24/P23/Q1(d)(ii)

Context: part (d) (introduction)

(d) The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, CaSiO_3 . The oxidation number of calcium in CaSiO_3 is +II.

(ii) Calcium oxide can be made from calcium carbonate in a single-step reaction.

Identify the type of reaction that occurs.

[1]

13. O/N 23/P21/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

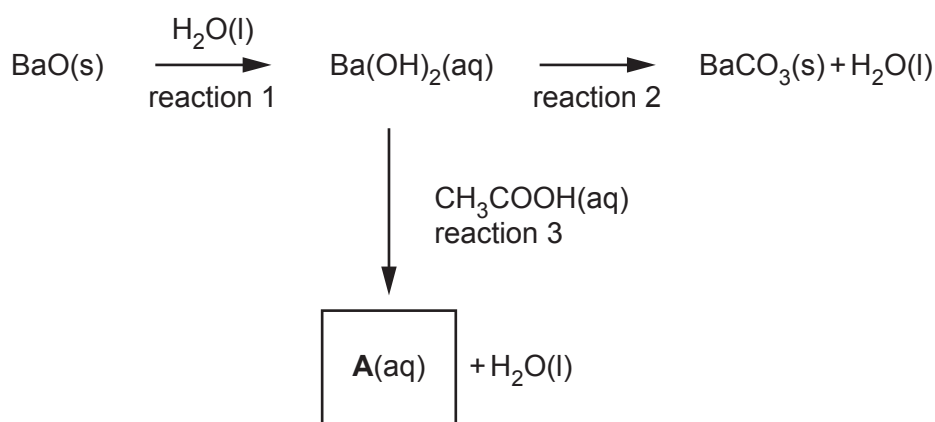


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

[1]

14. O/N 23/P21/Q2(b)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

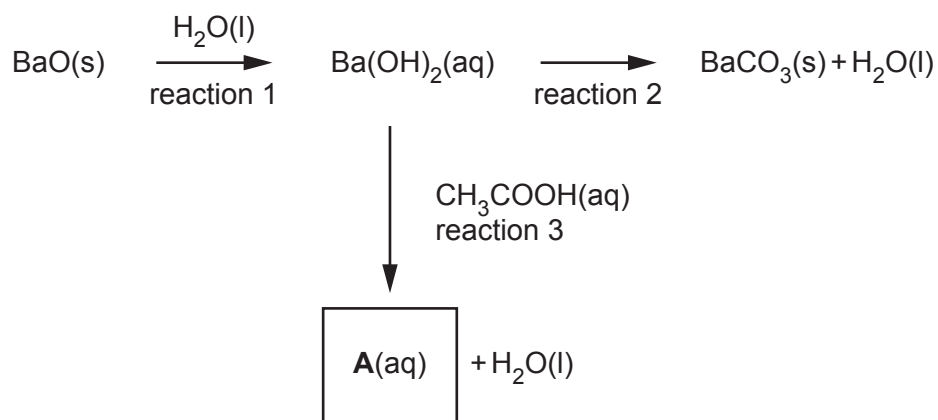


Fig. 2.1

(b) The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

(i) State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

(ii) Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

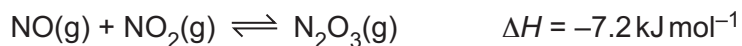
$\text{BaCa}(\text{CO}_3)_2$

[1]

15. O/N 23/P22/Q2(f)

Context

- 2** NO and NO₂ react at 25 °C to give N₂O₃ as shown in the equation.



The reaction is reversible and reaches equilibrium in a closed system.

- (f)** A student titrates nitric acid with a base to form a solution containing aqueous magnesium nitrate.

- (i)** Identify a base that the student could use.

[1]

- (ii)** The student evaporates the water to obtain magnesium nitrate solid. When this solid is heated it decomposes.

Write an equation for the decomposition of magnesium nitrate.

[1]

- (iii)** State how the thermal stability of Group 2 nitrates changes down the group.

[1]

16. O/N 23/P23/Q2(a)

Context

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

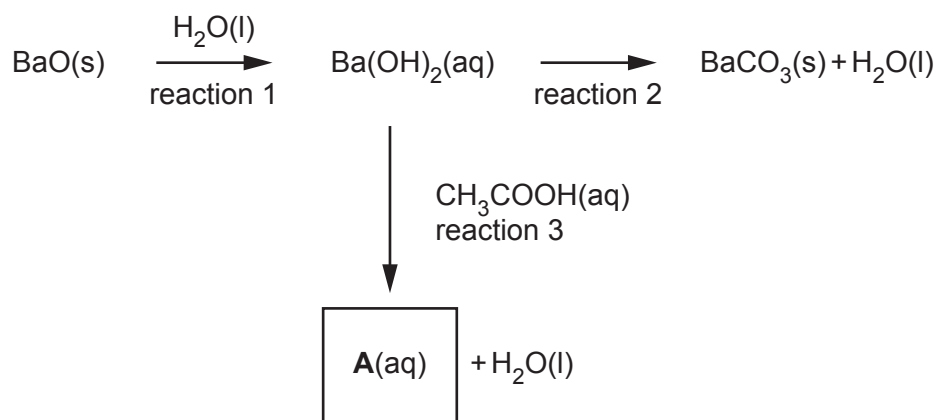


Fig. 2.1

(a) (i) State the variation in solubilities of group 2 hydroxides.

[1]

(ii) State what is observed in reaction 1.

[1]

(iii) Suggest a reactant for reaction 2.

[1]

(iv) Identify **A**.

[1]

(v) $\text{Ba}(\text{OH})_2$ is made by the reaction of Ba with water.

Write an equation for this reaction.

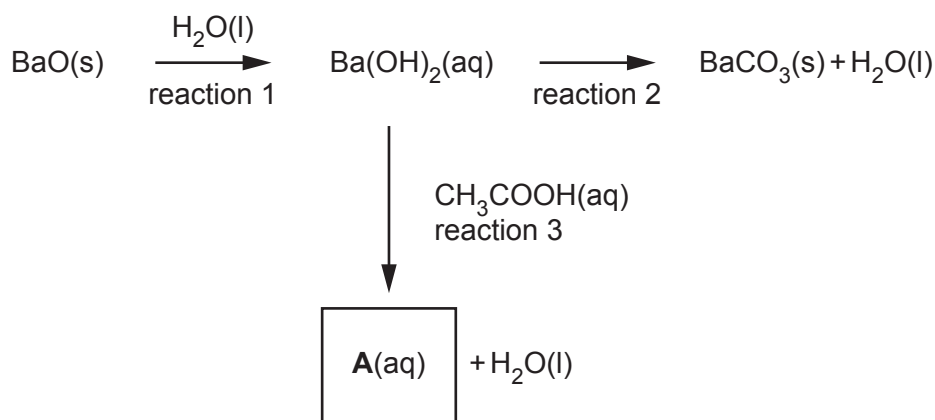
[1]

17. O/N 23/P23/Q2(b)

Context

- 2** Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

- (b)** The mineral barytocalcite contains both BaCO_3 and CaCO_3 . Both compounds decompose on heating.

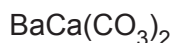
- (i)** State which compound decomposes first when barytocalcite is heated.

Explain your answer.

[1]

- (ii)** Construct an equation for the complete thermal decomposition of barytocalcite.

The formula of barytocalcite is $\text{BaCa}(\text{CO}_3)_2$.

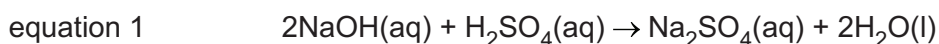


[1]

18. M/J 23/P21/Q2(c)(i)

Context

- 2** A neutralisation reaction occurs when NaOH(aq) is added to $\text{H}_2\text{SO}_4\text{(aq)}$.



- (c) (i)** Complete the equation for the reaction that occurs when a solution of $\text{Ba}(\text{OH})_2$ is added to aqueous sulfuric acid. Include state symbols.



[4]

19. M/J 23/P23/Q2(b)

- (b) Aluminium nitrate is a white soluble salt. On heating aluminium nitrate, thermal decomposition occurs and a brown gas is seen.

State the formula of the salt of another element in Period 3 which also decomposes on heating to produce a brown gas.

[1]

20. MAR 23/P22/Q1(d)(iii)

Context

- 1 The Pauling electronegativity values of elements can be used to predict the chemical properties of compounds.

Use the information in Table 1.1 to answer the following questions.

Table 1.1

element	H	Li	C	O	S
Pauling electronegativity value	2.1	1.0	2.5	3.5	2.6
first ionisation energy/kJ mol ⁻¹	1310	519	1090	1310	1000
second ionisation energy/kJ mol ⁻¹	—	7300	2350	3390	2260

Context: part (d) (introduction)

- (d) CO₂ and SO₂ are acidic gases.

(iii) Construct an equation for the reaction of CO₂ with Mg(OH)₂.

[1]

21. MAR 23/P22/Q2(b)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(b) Write an equation for the reaction of magnesium with cold water.

[1]

22. MAR 23/P22/Q2(c)

Context

- 2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(c) Identify a single reagent that can be used to distinguish separate samples of dilute Mg(NO₃)₂(aq) and dilute Ba(NO₃)₂(aq).

Explain your answer.

reagent

explanation

[2]

23. O/N 22/P21/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

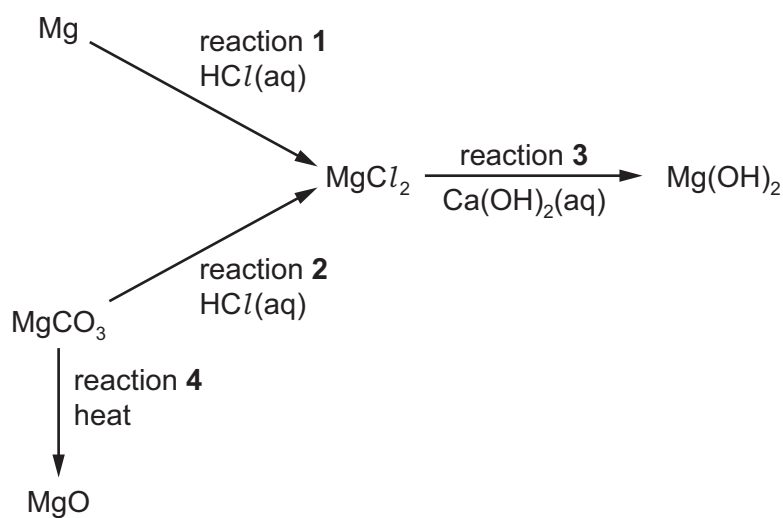


Fig. 2.1

(i) Identify the other products of reactions 1 and 2.

reaction 1

reaction 2

[2]

(ii) Reaction 3 is used to form a precipitate of Mg(OH)_2 from $\text{MgCl}_2\text{(aq)}$.

State why $\text{Ca(OH)}_2\text{(aq)}$ would **not** form a precipitate of Ba(OH)_2 from $\text{BaCl}_2\text{(aq)}$.

[1]

(iii) State the type of reaction that occurs in reaction 4.

[1]

24. O/N 22/P23/Q2(b)

Context

2 Magnesium shows reactions typical of a Group 2 metal.

(b) Fig. 2.1 shows some reactions of magnesium and its compounds.

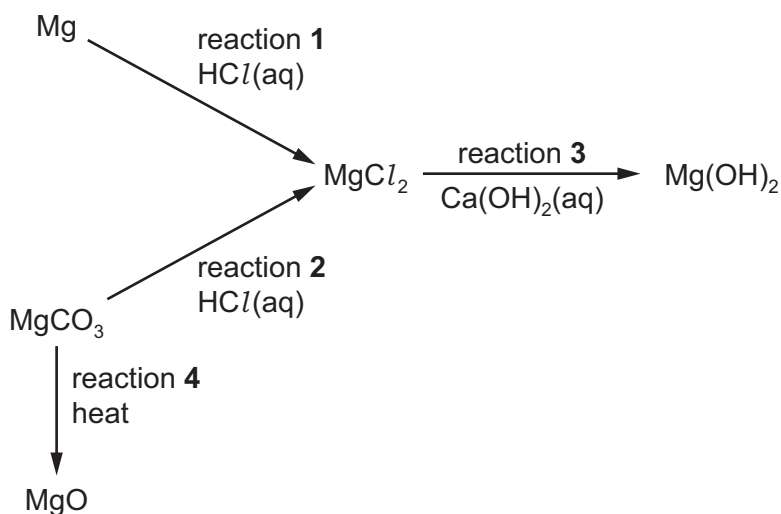


Fig. 2.1

(i) Identify the other products of reactions 1 and 2.

reaction 1

reaction 2

[2]

(ii) Reaction 3 is used to form a precipitate of $Mg(OH)_2$ from $MgCl_2(aq)$.

State why $Ca(OH)_2(aq)$ would **not** form a precipitate of $Ba(OH)_2$ from $BaCl_2(aq)$.

[1]

(iii) State the type of reaction that occurs in reaction 4.

[1]

25. M/J 22/P21/Q1(b)

Context

1 Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(b) (i) Write the equation for the thermal decomposition of calcium nitrate.

[1]

(ii) Suggest which of the Group 2 nitrates, calcium, magnesium or radium, requires the highest temperature to decompose. Explain your answer.

[1]

26. M/J 22/P21/Q1(c)

Context

- 1** Calcium, magnesium and radium are Group 2 elements. Radium follows the same trends as the other members of Group 2.

(c) Predict what you would observe when aqueous radium chloride is added to aqueous sodium sulfate.

Do **not** refer to temperature changes in your answer.

[1]

27. M/J 22/P22/Q2(c)

Context

- 2** Radium, Ra, is an element found in Group 2 of the Periodic Table. It is a crystalline solid at room temperature and conducts electricity.

Radium chloride, RaCl_2 , has a melting point of 900°C and is soluble in water.

(c) Solid Ra and Ca show similar reactions with H_2O , but the reactions occur at different rates.

Separate samples, each containing a single piece of solid Ra or Ca, are added to equal volumes of cold water.

Each sample contains equal numbers of moles of solid and the H_2O is in excess.

(i) Construct an equation for the reaction of Ra with H_2O .

[1]

(ii) Identify which element, Ra or Ca, reacts with H_2O at a faster rate. Suggest how the observations of each reaction would differ.

[1]

(iii) Suggest why these reactions occur at different rates.

[2]

(iv) One of the solutions is cloudy when the reaction has finished.

At the end of each reaction, universal indicator is added to each reaction mixture.

Suggest pH values of the solutions made in both reactions. Explain your answer.

[2]

Answers Section

1. M/J 26/P24/Q3(c)

3(c)(i)	brown gas	1	gas = fumes = vapour A red-brown gas, orange-brown gas A shades of brown R red gas (alone) I white solid (formed) I any other observations (e.g. melting, effervescence, solid disappears)
3(c)(ii)	increases	1	A decrease <u>up</u> the group. A more soluble

2. MAR 26/P22/Q1(b)

1(b)(i)	M1 $\text{Sr(s)} + 2\text{H}_2\text{O(l)} \rightarrow \text{Sr(OH)}_2\text{(aq)} + \text{H}_2\text{(g)}$ M2 state symbols for <u>all</u> species	2
1(b)(ii)	M1 with HCl no visible reaction / no change (observed) M2 with H_2SO_4 white precipitate/solid forms	2

3. MAR 26/P22/Q1(c)

1(c)(i)	(thermal stability) increases (down the group)	1
1(c)(ii)	$2\text{Ca(NO}_3)_2 \rightarrow 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$	1

4. O/N 25/P21/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

5. O/N 25/P22/Q2(f)

2(f)(i)	$\text{MgCl}_2\text{(aq)} + \text{Na}_2\text{CO}_3\text{(aq)} \rightarrow \text{MgCO}_3\text{(s)} + 2\text{NaCl(aq)}$ M1 formulae in balanced equation M2 state symbols	2
2(f)(ii)	MgO / magnesium oxide AND CO_2 / carbon dioxide	1
2(f)(iii)	(thermal stability) increases (down the group)	1

6. O/N 25/P23/Q2(d)

2(d)(i)	(thermal stability) increases (down the group)	1
2(d)(ii)	NO_2 AND O_2	1

7. MAR 25/P22/Q2(d)

2(d)(i)	<i>barium oxide</i> (white) solid disappears / dissolves <i>barium carbonate</i> effervescence AND (white) solid disappears / dissolves	2
2(d)(ii)	$\text{Sr} + 2\text{H}_2\text{O} \rightarrow \text{Sr(OH)}_2 + \text{H}_2$	1
2(d)(iii)	decreases (down the group)	1

8. O/N 24/P21/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

9. O/N 24/P22/Q3(a)(iv)-(v)

3(a)(iv)	$2\text{HNO}_3 + \text{CaO} \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O}$	1
3(a)(v)	brown fumes given off	1

10. O/N 24/P23/Q3(b)(i)-(ii)

3(b)(i)	$\text{MgCO}_3 + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$	1
3(b)(ii)	thermal stability increases down the group	1

11. M/J 24/P22/Q2(c)

2(c)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$ M1 correct formula for calcium nitrate: $\text{Ca}(\text{NO}_3)_2$ M2 correct balanced equation	2
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12. M/J 24/P23/Q1(d)(ii)

1(d)(ii)	thermal decomposition	1
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13. O/N 23/P21/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1
2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1

14. O/N 23/P21/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

15. O/N 23/P22/Q2(f)

2(f)(i)	magnesium hydroxide / $\text{Mg}(\text{OH})_2$	1
2(f)(ii)	$\text{Mg}(\text{NO}_3)_2 \rightarrow \text{MgO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
2(f)(iii)	increases	1

16. O/N 23/P23/Q2(a)

2(a)(i)	increases down the group	1
2(a)(ii)	white solid disappears	1
2(a)(iii)	CO_2 / carbon dioxide	1
2(a)(iv)	$\text{Ba}(\text{CH}_3\text{COO})_2$ / barium ethanoate	1

2(a)(v)	$\text{Ba} + 2\text{H}_2\text{O} \rightarrow \text{Ba}(\text{OH})_2 + \text{H}_2$	1
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17. O/N 23/P23/Q2(b)

2(b)(i)	CaCO_3 AND thermal stability of carbonates increases down the group	1
2(b)(ii)	$\text{BaCa}(\text{CO}_3)_2 \rightarrow \text{BaO} + \text{CaO} + 2\text{CO}_2$	1

18. M/J 23/P21/Q2(c)(i)

2(c)(i)	..(1)..$\text{H}_2\text{SO}_4(\text{aq}) + \text{..(1)..Ba}(\text{OH})_2(\text{aq}) \rightarrow \text{...(1)..BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$..... M1 correct state symbols for both products M2 correctly balanced equation	2
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19. M/J 23/P23/Q2(b)

2(b)	$\text{Mg}(\text{NO}_3)_2$	1
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20. MAR 23/P22/Q1(d)(iii)

1(d)(iii)	$\text{CO}_2 + \text{Mg}(\text{OH})_2 \rightarrow \text{MgCO}_3 + \text{H}_2\text{O}$	1
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21. MAR 23/P22/Q2(b)

2(b)	$\text{Mg} + 2\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{H}_2$	1
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22. MAR 23/P22/Q2(c)

2(c)	reagent = any named/formula of soluble sulfate OR H_2SO_4 OR any named/formula of soluble hydroxide	1
	BaSO₄ insoluble (& MgSO₄ soluble) OR (Ba(OH)₂ soluble &) Mg(OH)₂ insoluble	1

23. O/N 22/P21/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

24. O/N 22/P23/Q2(b)

2(b)(i)	reaction 1 = hydrogen / H_2	1
	reaction 2 = carbon dioxide / CO_2 AND water / H_2O	1
2(b)(ii)	$\text{Ba}(\text{OH})_2$ is soluble (in aqueous solution) / solubility of Group 2 hydroxides increases down group	1
2(b)(iii)	thermal decomposition	1

25. M/J 22/P21/Q1(b)

1(b)(i)	$\text{Ca}(\text{NO}_3)_2 \rightarrow \text{CaO} + 2\text{NO}_2 + \frac{1}{2}\text{O}_2$	1
1(b)(ii)	radium (nitrate) as thermal stability increases down group / has the greatest thermal stability	1

26. M/J 22/P21/Q1(c)

1(c)	white precipitate / solid (of radium sulfate)	1
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27. M/J 22/P22/Q2(c)

2(c)(i)	$\text{Ra} + 2\text{H}_2\text{O} \rightarrow \text{Ra}(\text{OH})_2 + \text{H}_2$	1
2(c)(ii)	Ra – more bubbles per unit time OR With Ra solid disappears more quickly OR Ra is the first to stop fizzing ora	1
2(c)(iii)	option 1 <i>suggest why these reactions occur at different rates (collision theory)</i> M1 difference in activation energy / ionisation energy M2 affects the frequency of effective collisions (between particles / molecules) OR affects the proportion of particles with energy greater than activation energy option 2 <i>suggest why the elements have different reactivity in terms of atomic structure (which results in a different rate)</i> M1 ionisation energies are different M2 a decrease in nuclear attraction due to EITHER increase in shielding OR increase in distance of outer electron from nucleus OR increase in number of shells of electrons ora	2
2(c)(iv)	M1 - pH value (or values) for each solution must be greater > 7 AND - pH value (or range values) identified for solution made from Ra must be greater than pH values (or range of values) stated for Ca M2 any one from: - solubility of group 2 hydroxides increases down the group $\text{Ra}(\text{OH})_2$ is more soluble - greater concentration of $\text{OH}^-(\text{aq})$ in the solution involving Ra - more $\text{OH}^-(\text{aq})$ in the solution involving Ra	2